

## Creating python backed without using docker:

```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS

PS C:\Users\hp\Downloads\python-backend\python-backend> python -m venv venv
• >> venv\Scripts\Activate.ps1
• (venv) PS C:\Users\hp\Downloads\python-backend\python-backend> pip install -r requirements.txt
Collecting fastapi (from -r requirements.txt (line 1))
  Downloading fastapi-0.139.2-py3-none-any.whl.metadata (26 kB)
Collecting uvicorn (from -r requirements.txt (line 2))
  Downloading uvicorn-0.51.0-py3-none-any.whl.metadata (6.6 kB)
Collecting starlette>=0.46.0 (from fastapi->-r requirements.txt (line 1))
  Downloading starlette-1.3.1-py3-none-any.whl.metadata (6.4 kB)
Collecting pydantic>=2.9.0 (from fastapi->-r requirements.txt (line 1))
  Downloading pydantic-2.13.4-py3-none-any.whl.metadata (109 kB)
Collecting typing-extensions>=4.8.0 (from fastapi->-r requirements.txt (line 1))
  Downloading typing_extensions-4.16.0-py3-none-any.whl.metadata (3.3 kB)
Collecting typing-inspection>=0.4.2 (from fastapi->-r requirements.txt (line 1))
  Downloading typing_inspection-0.4.2-py3-none-any.whl.metadata (2.6 kB)
Collecting annotated-doc>=0.0.2 (from fastapi->-r requirements.txt (line 1))
  Downloading annotated_doc-0.0.4-py3-none-any.whl.metadata (6.6 kB)
Collecting click>=7.0 (from uvicorn->-r requirements.txt (line 2))
  Downloading click-8.4.2-py3-none-any.whl.metadata (2.6 kB)
Collecting h11>=0.8 (from uvicorn->-r requirements.txt (line 2))
  Downloading h11-0.16.0-py3-none-any.whl.metadata (8.3 kB)
Collecting colorama (from click>=7.0->uvicorn->-r requirements.txt (line 2))
  Downloading colorama-0.4.6-py2.py3-none-any.whl.metadata (17 kB)
Collecting annotated-types>=0.6.0 (from pydantic>=2.9.0->fastapi->-r requirements.txt (line 1))
  Downloading annotated_types-0.7.0-py3-none-any.whl.metadata (15 kB)
Collecting pydantic-core==2.46.4 (from pydantic>=2.9.0->fastapi->-r requirements.txt (line 1))
  Downloading pydantic_core-2.46.4-cp314-cp314-win_amd64.whl.metadata (6.7 kB)

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PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS

(venv) PS C:\Users\hp\Downloads\python-backend\python-backend> pip install -r requirements.txt
2.1/2.1 MB 1.0 MB/s 0:00:02
Downloading annotated_types-0.7.0-py3-none-any.whl (13 kB)
Downloading starlette-1.3.1-py3-none-any.whl (73 kB)
Downloading anyio-4.14.2-py3-none-any.whl (125 kB)
Downloading idna-3.18-py3-none-any.whl (65 kB)
Downloading typing_extensions-4.16.0-py3-none-any.whl (45 kB)
Downloading typing_inspection-0.4.2-py3-none-any.whl (14 kB)
Downloading colorama-0.4.6-py2.py3-none-any.whl (25 kB)
Installing collected packages: typing-extensions, idna, h11, colorama, annotated-types, annotated-doc, typing-inspection, pydantic-core, click, anyio, uvicorn, starlette, pydantic, fastapi
Successfully installed annotated-doc-0.0.4 annotated-types-0.7.0 anyio-4.14.2 click-8.4.2 colorama-0.4.6 fastapi-0.139.2 h11-0.16.0 idna-3.18 pydantic-2.13.4 pydantic-core-2.46.4 starlette-1.3.1 typing-extensions-4.16.0 typing-inspection-0.4.2 uvicorn-0.51.0

[notice] A new release of pip is available: 25.3 -> 26.1.2
[notice] To update, run: python.exe -m pip install --upgrade pip
(venv) PS C:\Users\hp\Downloads\python-backend\python-backend> uvicorn main:app --reload
INFO: Will watch for changes in these directories: ['C:\Users\hp\Downloads\python-backend\python-backend']
INFO: Uvicorn running on http://127.0.0.1:8000 (Press CTRL+C to quit)
INFO: Started reloader process [5664] using StatReload
INFO: Started server process [21388]
INFO: Waiting for application startup.
INFO: Application startup complete.
INFO: 127.0.0.1:63864 - "GET / HTTP/1.1" 200 OK
INFO: 127.0.0.1:63864 - "GET /favicon.ico HTTP/1.1" 404 Not Found
[]

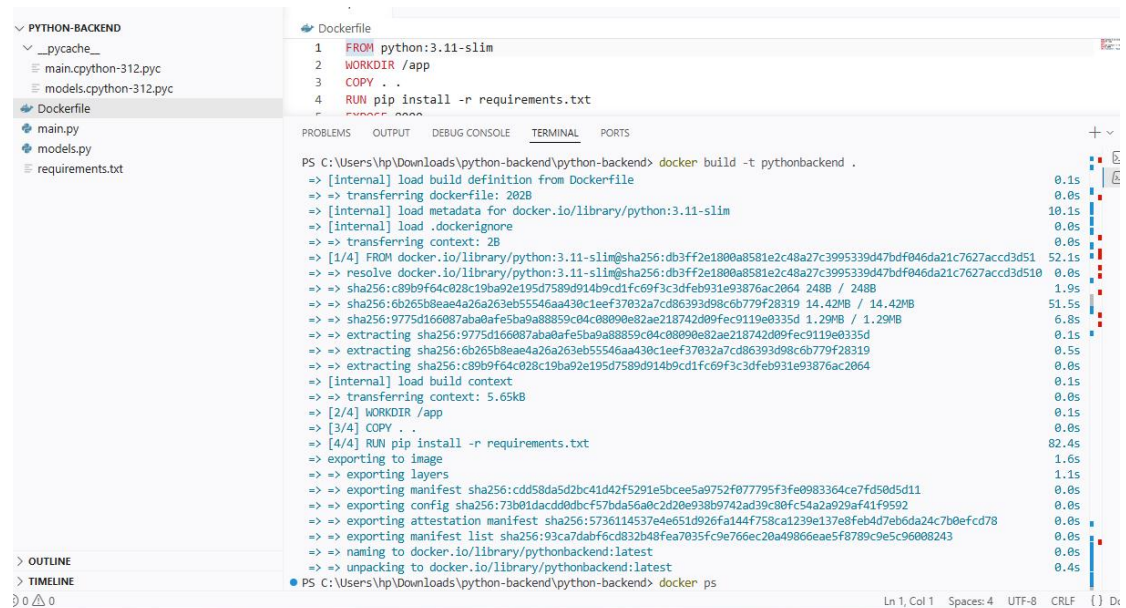
← → ↺ ⓘ localhost:8000

🗖 | 📧 Gmail 📺 YouTube 📍 Maps 🏠 Console Home | Co... ☁ AWS Sessions SRE 5...

Pretty print ☐

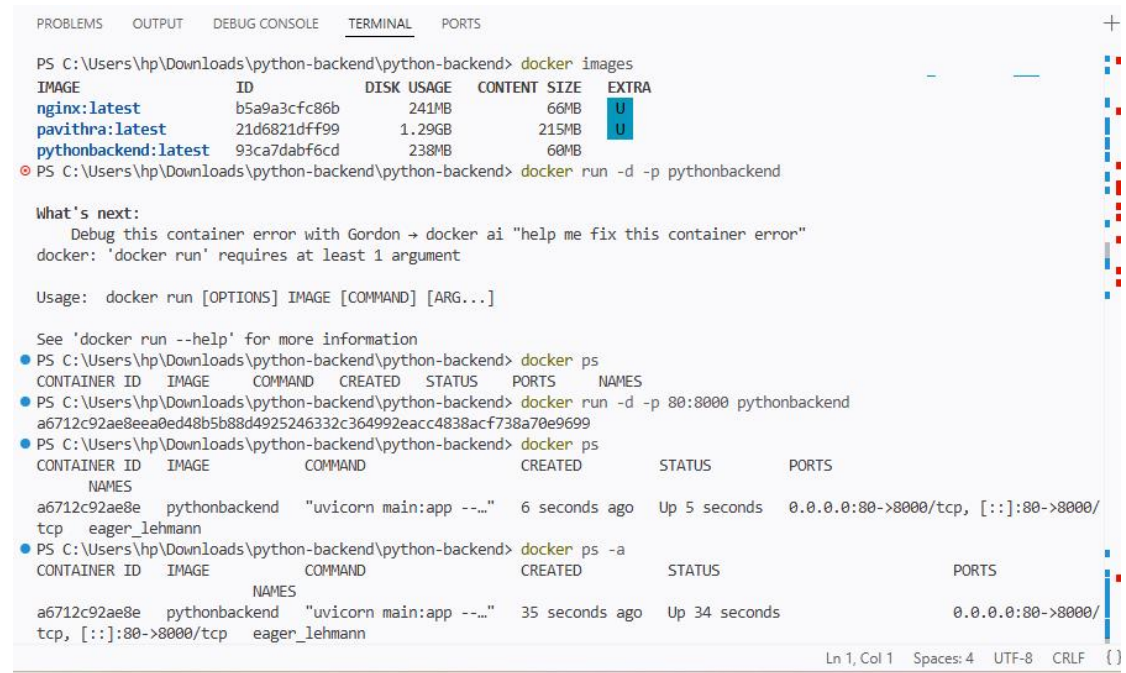
"Hello from python backend server"
```

## Creating pythonbackend using docker:



```
Dockerfile
1 FROM python:3.11-slim
2 WORKDIR /app
3 COPY . .
4 RUN pip install -r requirements.txt

PS C:\Users\hp\Downloads\python-backend\python-backend> docker build -t pythonbackend .
=> [internal] load build definition from Dockerfile
=> => transferring dockerfile: 202B
=> [internal] load metadata for docker.io/library/python:3.11-slim
=> [internal] load .dockerignore
=> => transferring context: 2B
=> [1/4] FROM docker.io/library/python:3.11-slim@sha256:db3ff2e1800a8581e2c48a27c3995339d47bdf046da21c7627accd3d51
=> => resolve docker.io/library/python:3.11-slim@sha256:db3ff2e1800a8581e2c48a27c3995339d47bdf046da21c7627accd3d51
=> => sha256:c89b9f64c828c19ba92e195d7589d914b9cd1fc69f3c3dfeb931e93876ac2064 2488 / 2488
=> => sha256:6b265b8eae4a26a263eb5546aa430c1ee3f7032a7cd86393d98c6b779f28319 14.42MB / 14.42MB
=> => sha256:9775d166087aba0afe5ba9a88859c04c08090e82ae218742d09fec9119e0335d 1.29MB / 1.29MB
=> => extracting sha256:6b265b8eae4a26a263eb5546aa430c1ee3f7032a7cd86393d98c6b779f28319
=> => extracting sha256:c89b9f64c828c19ba92e195d7589d914b9cd1fc69f3c3dfeb931e93876ac2064
=> [internal] load build context
=> => transferring context: 5.65kB
=> [2/4] WORKDIR /app
=> [3/4] COPY . .
=> [4/4] RUN pip install -r requirements.txt
=> exporting to image
=> => exporting layers
=> => exporting manifest sha256:cdd58da52bc41d42f5291e5bcee5a9752f07795f3fe9803364ce7fd50d5d11
=> => exporting config sha256:73b01dacdd8dbcf57bda56a0c2d20e93809742ad39c80fc54a2a929af41f9592
=> => exporting attestation manifest sha256:5736114537e4e651d826fa144f758ca1239e137e8feb4d7eb6da24c7b0efcd78
=> => exporting manifest list sha256:93ca7dabf6cd832b48fea7035fc9e766ec2ba49866eae5f8789c9e5c96008243
=> => naming to docker.io/library/pythonbackend:latest
=> => unpacking to docker.io/library/pythonbackend:latest
PS C:\Users\hp\Downloads\python-backend\python-backend> docker ps
```



```
PS C:\Users\hp\Downloads\python-backend\python-backend> docker images
IMAGE ID DISK USAGE CONTENT SIZE EXTRA
nginx:latest b5a9a3cfc86b 241MB 66MB U
pavithra:latest 21d6821dff99 1.29GB 215MB U
pythonbackend:latest 93ca7dabf6cd 238MB 60MB

PS C:\Users\hp\Downloads\python-backend\python-backend> docker run -d -p pythonbackend

What's next:
Debug this container error with Gordon → docker ai "help me fix this container error"
docker: 'docker run' requires at least 1 argument

Usage: docker run [OPTIONS] IMAGE [COMMAND] [ARG...]

See 'docker run --help' for more information
PS C:\Users\hp\Downloads\python-backend\python-backend> docker ps
CONTAINER ID IMAGE COMMAND CREATED STATUS PORTS NAMES
a6712c92ae8e pythonbackend "uvicorn main:app --..." 6 seconds ago Up 5 seconds 0.0.0.0:80->8000/tcp, [::]:80->8000/tcp eager_lehmann
PS C:\Users\hp\Downloads\python-backend\python-backend> docker run -d -p 80:8000 pythonbackend
a6712c92ae8e pythonbackend "uvicorn main:app --..." 6 seconds ago Up 5 seconds 0.0.0.0:80->8000/tcp, [::]:80->8000/tcp eager_lehmann
PS C:\Users\hp\Downloads\python-backend\python-backend> docker ps
CONTAINER ID IMAGE COMMAND CREATED STATUS PORTS NAMES
a6712c92ae8e pythonbackend "uvicorn main:app --..." 35 seconds ago Up 34 seconds 0.0.0.0:80->8000/tcp, [::]:80->8000/tcp eager_lehmann
```



```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
PS C:\Users\hp\Downloads\python-backend\python-backend> docker ps -a
CONTAINER ID IMAGE COMMAND CREATED STATUS PORTS
a6712c92ae8e pythonbackend "uvicorn main:app --..." 35 seconds ago Up 34 seconds 0.0.0.0:80->8000/tcp, [::]:80->8000/tcp
b4deba3a59eb eager_lehmann "docker-entrypoint.s..." 34 hours ago Exited (255) About an hour ago 0.0.0.0:3000->3000/tcp
2b73142a48c8 nginx "/docker-entrypoint..." 3 days ago Exited (255) 35 hours ago 0.0.0.0:8080->8000/tcp
PS C:\Users\hp\Downloads\python-backend\python-backend> docker stop a6712c92ae8e
a6712c92ae8e
PS C:\Users\hp\Downloads\python-backend\python-backend> docker ps
CONTAINER ID IMAGE COMMAND CREATED STATUS PORTS NAMES
b4deba3a59eb pavithra "docker-entrypoint.s..." 34 hours ago Exited (255) About an hour ago 0.0.0.0:3000->3000/tcp
2b73142a48c8 nginx "/docker-entrypoint..." 3 days ago Exited (255) 35 hours ago 0.0.0.0:8080->8000/tcp
PS C:\Users\hp\Downloads\python-backend\python-backend> docker rm a6712c92ae8e
a6712c92ae8e
```

```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
PS C:\Users\hp\Downloads\python-backend\python-backend> docker ps -a
CONTAINER ID IMAGE COMMAND CREATED STATUS PORTS
b4deba3a59eb pavithra "docker-entrypoint.s..." 34 hours ago Exited (255) About an hour ago 0.0.0.0:3000->3000/tcp
2b73142a48c8 nginx "/docker-entrypoint..." 3 days ago Exited (255) 35 hours ago 0.0.0.0:8080->8000/tcp
PS C:\Users\hp\Downloads\python-backend\python-backend> docker run --name python container -d -p 80:8000 pythonbackend
Unable to find image 'container:latest' locally

What's next:
Debug this container error with Gordon -> docker ai "help me fix this container error"
docker: Error response from daemon: pull access denied for container, repository does not exist or may require 'docker login'

Run 'docker run --help' for more information
PS C:\Users\hp\Downloads\python-backend\python-backend> docker run --name python-container -d -p 80:8000 pythonbackend
3b2c1feab7b976744246854bf0b2dec21b6a4c953bf0cb34954759f621ef4662
PS C:\Users\hp\Downloads\python-backend\python-backend> docker ps
CONTAINER ID IMAGE COMMAND CREATED STATUS PORTS
3b2c1feab7b9 pythonbackend "uvicorn main:app --..." 13 seconds ago Up 12 seconds 0.0.0.0:80->8000/tcp, [::]:80->8000/tcp
0/tcp python-container
```

```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
PS C:\Users\hp\Downloads\python-backend\python-backend> docker exec -it pythonbackend sh

What's next:
Try Docker Debug for seamless, persistent debugging tools in any container or image -> docker debug pythonbackend
Learn more at https://docs.docker.com/go/debug-cli/
Error response from daemon: No such container: pythonbackend
PS C:\Users\hp\Downloads\python-backend\python-backend> docker exec python-container sh
PS C:\Users\hp\Downloads\python-backend\python-backend> docker exec -it python-container sh
# pwd
/app
# cd
# pwd
/root
# cd..
sh: 4: cd.: not found
# exit

What's next:
Try Docker Debug for seamless, persistent debugging tools in any container or image -> docker debug python-container
Learn more at https://docs.docker.com/go/debug-cli/
PS C:\Users\hp\Downloads\python-backend\python-backend> docker exec -it pythonbackend

What's next:
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Learn more at https://docs.docker.com/go/debug-cli/
docker: 'docker exec' requires at least 2 arguments

Usage: docker exec [OPTIONS] CONTAINER COMMAND [ARG...]
```

```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
PS C:\Users\hp\Downloads\python-backend\python-backend> docker exec -it pythonbackend sh
PS C:\Users\hp\Downloads\python-backend\python-backend> docker exec -it python-container sh
# pwd
/app
# cd ..
# pwd
/
# cd /app
# ls
Dockerfile __pycache__ main.py models.py requirements.txt
# exit

What's next:
Try Docker Debug for seamless, persistent debugging tools in any container or image → docker debug python-container
Learn more at https://docs.docker.com/go/debug-cli/
PS C:\Users\hp\Downloads\python-backend\python-backend> docker logs python-container sh
docker: 'docker logs' requires 1 argument

Usage: docker logs [OPTIONS] CONTAINER

Run 'docker logs --help' for more information
PS C:\Users\hp\Downloads\python-backend\python-backend> ^C
```

```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
PS C:\Users\hp\Downloads\python-backend\python-backend> docker logs --tail 2 python-container
INFO: Application startup complete.
INFO: Uvicorn running on http://0.0.0.0:8000 (Press CTRL+C to quit)
PS C:\Users\hp\Downloads\python-backend\python-backend> docker logs python-container
INFO: Started server process [1]
INFO: Waiting for application startup.
INFO: Application startup complete.
INFO: Uvicorn running on http://0.0.0.0:8000 (Press CTRL+C to quit)
PS C:\Users\hp\Downloads\python-backend\python-backend> docker logs -f python-container
INFO: Started server process [1]
INFO: Waiting for application startup.
INFO: Application startup complete.
INFO: Uvicorn running on http://0.0.0.0:8000 (Press CTRL+C to quit)
INFO: 172.17.0.1:33390 - "GET / HTTP/1.1" 200 OK
INFO: 172.17.0.1:33390 - "GET /favicon.ico HTTP/1.1" 404 Not Found
INFO: 172.17.0.1:33402 - "GET / HTTP/1.1" 200 OK
INFO: 172.17.0.1:45446 - "GET / HTTP/1.1" 200 OK
INFO: 172.17.0.1:45446 - "GET / HTTP/1.1" 200 OK
INFO: 172.17.0.1:37188 - "GET / HTTP/1.1" 200 OK
PS C:\Users\hp\Downloads\python-backend\python-backend> docker logs -f python-container
INFO: Started server process [1]
INFO: Waiting for application startup.
INFO: Application startup complete.
INFO: Uvicorn running on http://0.0.0.0:8000 (Press CTRL+C to quit)
INFO: 172.17.0.1:33390 - "GET / HTTP/1.1" 200 OK
INFO: 172.17.0.1:33390 - "GET /favicon.ico HTTP/1.1" 404 Not Found
INFO: 172.17.0.1:33402 - "GET / HTTP/1.1" 200 OK
INFO: 172.17.0.1:45446 - "GET / HTTP/1.1" 200 OK
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```

```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
PS C:\Users\hp\Downloads\python-backend\python-backend> docker inspect python-container
"8000"
},
"Image": "pythonbackend",
"Volumes": null,
"WorkingDir": "/app",
"Entrypoint": null,
"Labels": {}
},
"NetworkSettings": {
  "SandboxID": "4d27c105d2a341f642b310866dd4bcff80a23ad5aa9d5555df9683356d2b6566",
  "SandboxKey": "/var/run/docker/netns/4d27c105d2a3",
  "Ports": {
    "8000/tcp": [
      {
        "HostIp": "0.0.0.0",
        "HostPort": "80"
      }
    ]
  }
}
```



PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS C:\Users\hp\Downloads\python-backend\python-backend> docker inspect python-container
[{"Status": "running",
  "Running": true,
  "Paused": false,
  "Restarting": false,
  "OOMKilled": false,
  "Dead": false,
  "Pid": 598,
  "ExitCode": 0,
  "Error": "",
  "StartedAt": "2026-07-18T09:15:28.559617129Z",
  "FinishedAt": "0001-01-01T00:00:00Z"},
{"Image": "sha256:93ca7dabf6cd832b48fea7035fc9e766ec20a49866eae5f8789c9e5c96008243",
  "ResolvConfPath": "/var/lib/docker/containers/3b2c1feab7b976744246854bf0b2dec21b6a4c953bf0cb34954759f621ef4662/resolv.conf",
  "HostnamePath": "/var/lib/docker/containers/3b2c1feab7b976744246854bf0b2dec21b6a4c953bf0cb34954759f621ef4662/hostname",
  "HostsPath": "/var/lib/docker/containers/3b2c1feab7b976744246854bf0b2dec21b6a4c953bf0cb34954759f621ef4662/hosts",
  "LogPath": "/var/lib/docker/containers/3b2c1feab7b976744246854bf0b2dec21b6a4c953bf0cb34954759f621ef4662/3b2c1feab7b976744246854bf0b2dec21b6a4c953bf0cb34954759f621ef4662-json.log",
  "Name": "/python-container",
  "RestartCount": 0,
  "Driver": "overlayfs",
  "Platform": "linux",
  "MountLabel": "",
  "ProcessLabel": ""}]
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS C:\Users\hp\Downloads\python-backend\python-backend> docker exec -it python-container sh
# ls
Dockerfile __pycache__ main.py models.py requirements.txt
# mkdir pavi
# ls
Dockerfile __pycache__ main.py models.py pavi requirements.txt
# cd pavi
# echo "hello pavi">text.txt
# ls
text.txt
# cat text.txt
hello pavi
# exit

What's next:
Try Docker Debug for seamless, persistent debugging tools in any container or image → docker debug python-container
Learn more at https://docs.docker.com/go/debug-cli/

PS C:\Users\hp\Downloads\python-backend\python-backend> docker ps
CONTAINER ID   IMAGE          COMMAND                  CREATED        STATUS        PORTS        NAMES
3b2c1feab7b9   pythonbackend  "uvicorn main:app --..." 15 minutes ago Up 15 minutes 0.0.0.0:80->8000/tcp, [::]:80->8000/tcp   python-container

PS C:\Users\hp\Downloads\python-backend\python-backend> docker rm python-container
Error response from daemon: cannot remove container "python-container": container is running: stop the container before removing or force remove

PS C:\Users\hp\Downloads\python-backend\python-backend> docker rm -f python-container
python-container

PS C:\Users\hp\Downloads\python-backend\python-backend> docker ps
CONTAINER ID   IMAGE          COMMAND                  CREATED        STATUS        PORTS        NAMES
```

```
PS C:\Users\hp\Downloads\python-backend\python-backend> docker exec -it python-container sh
PS C:\Users\hp\Downloads\python-backend\python-backend> docker ps
CONTAINER ID   IMAGE          COMMAND                  CREATED        STATUS        PORTS        NAMES
3b2c1feab7b9   pythonbackend  "uvicorn main:app --..." 15 minutes ago Up 15 minutes 0.0.0.0:80->8000/tcp, [::]:80->8000/tcp   python-container

PS C:\Users\hp\Downloads\python-backend\python-backend> docker rm python-container
Error response from daemon: cannot remove container "python-container": container is running: stop the container before removing or force remove

PS C:\Users\hp\Downloads\python-backend\python-backend> docker rm -f python-container
python-container

PS C:\Users\hp\Downloads\python-backend\python-backend> docker ps
CONTAINER ID   IMAGE          COMMAND                  CREATED        STATUS        PORTS        NAMES
PS C:\Users\hp\Downloads\python-backend\python-backend> docker run --name python-container -d -p 80:8000 pythonbackend
7b172ffe15bd4378b6a688edda0e0364544c1fe0113e76b010ba94c1aacc7fe8
PS C:\Users\hp\Downloads\python-backend\python-backend> docker ps
CONTAINER ID   IMAGE          COMMAND                  CREATED        STATUS        PORTS        NAMES
7b172ffe15bd   pythonbackend  "uvicorn main:app --..." 6 seconds ago  Up 5 seconds  0.0.0.0:80->8000/tcp, [::]:80->8000/tcp   python-container
```

```
PS C:\Users\hp\Downloads\python-backend\python-backend> docker ps -a
CONTAINER ID   IMAGE          COMMAND                  CREATED        STATUS        PORTS                    NAME
b4deba3a59eb   pavithra      "docker-entrypoint.s..." 35 hours ago   Exited (255) 2 hours ago   0.0.0.0:3000->3000/tcp   focused_clarke
2b73142a48c8   nginx        "/docker-entrypoint..." 3 days ago     Exited (255) 35 hours ago   0.0.0.0:8080->80/tcp     epic_spence

PS C:\Users\hp\Downloads\python-backend\python-backend> docker run --name python-container -d -p 80:8000 pythonbackend
ba993a52daef3024085ab097ebc7f2dc9291edb01e3893446acd59bdb2d77236

PS C:\Users\hp\Downloads\python-backend\python-backend> docker ps
CONTAINER ID   IMAGE          COMMAND                  CREATED        STATUS        PORTS                    NAMES
ba993a52daef   pythonbackend "uvicorn main:app --..." 5 seconds ago   Up 5 seconds   0.0.0.0:80->8000/tcp, [::]:80->8000/tcp   python-container

PS C:\Users\hp\Downloads\python-backend\python-backend> docker stop ba993a52daef
ba993a52daef

PS C:\Users\hp\Downloads\python-backend\python-backend> docker ps
CONTAINER ID   IMAGE          COMMAND                  CREATED        STATUS        PORTS                    NAMES

PS C:\Users\hp\Downloads\python-backend\python-backend> docker images

INFO → In Use
IMAGE          ID                DISK USAGE  CONTENT SIZE  EXTRA
nginx:latest   b5a9a3cfc86b     241MB       66MB          U
pavithra:latest 21d6821dff99     1.29GB     215MB          U
pythonbackend:latest 93ca7dabf6cd    238MB       60MB          U

PS C:\Users\hp\Downloads\python-backend\python-backend>
```

